

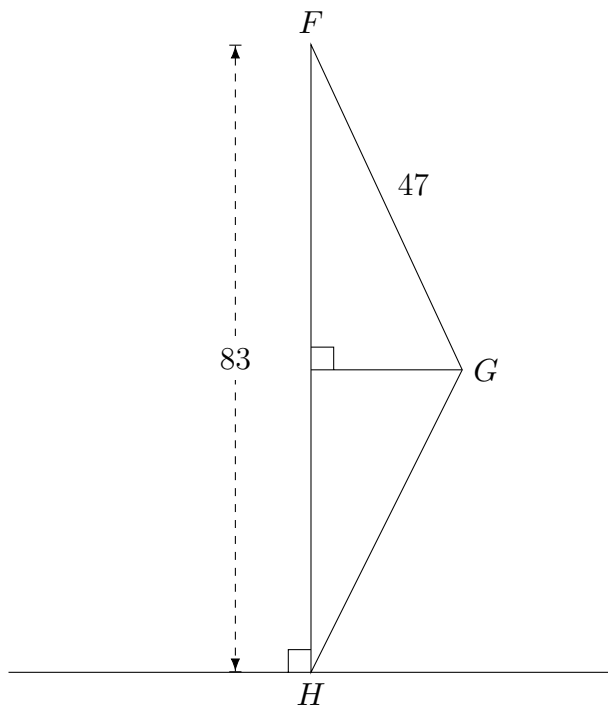
## Standard 2 Trigonometry

### Question Bank

#### Question 1

The triangle  $FGH$  is shown below. The length of  $FG$  is 47 and the length of  $FH$  is 83.

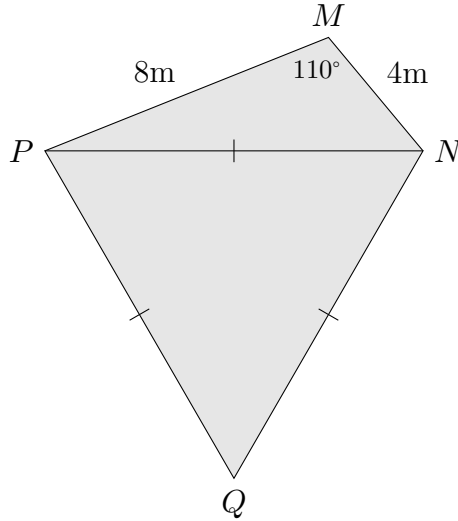
The angle of elevation of  $G$  from  $H$  is  $64^\circ$  and  $\angle FGH$  is obtuse.



Find the length of  $GH$  correct to two decimal places.

### Question 2

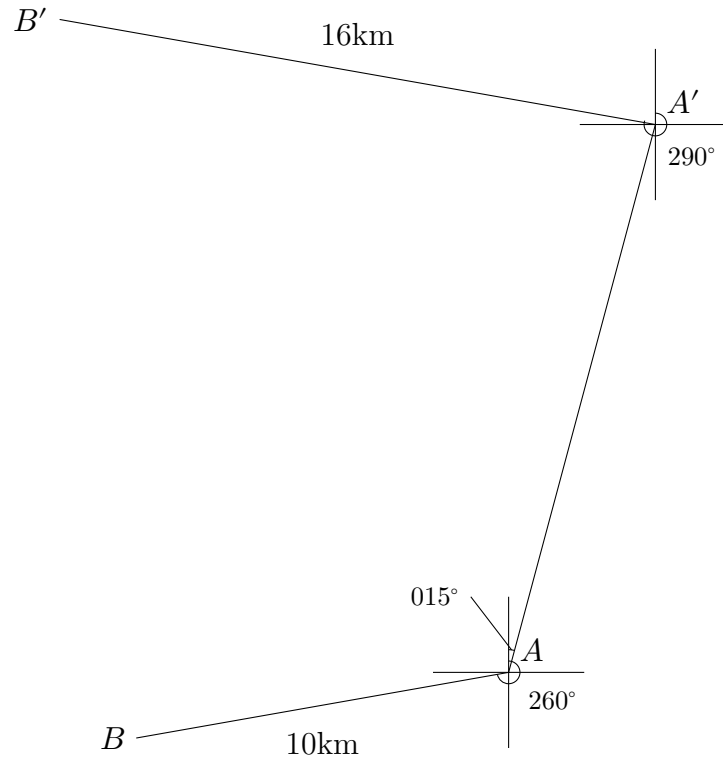
A strange window is constructed from two triangles,  $PMN$  and  $PNQ$ . The length of  $PM$  is 8m and the length of  $MN$  is 4m. The  $\angle PMN$  is  $110^\circ$ .



The window is to be coated in a protective film that costs  $\$45/\text{m}^2$ , how much will it cost to coat the window?

### Question 3

When Ship 1 is location at position  $A$  it determines that Ship 2 is 10km away on a bearing of  $260^\circ$  at point  $B$ . Then Ship 1 sails at a speed of 5km/hr on a bearing of  $15^\circ$  for 3 hours, reaching the point  $A'$ . Then it made another measurement and found that Ship 2 was 16km away on a bearing of  $290^\circ$  at point  $B'$ .



How fast is Ship 2 moving? (Extension question is to find the bearing Ship 2 is travelling on).

## Answers

### Question 1

Since  $\angle FHG$  is complementary to  $64^\circ$ ,

$$\angle FHG = 26^\circ.$$

Find  $\angle FGH$  using sine rule,

$$\begin{aligned}\frac{\sin(\angle FGH)}{83} &= \frac{\sin(26^\circ)}{47} \\ \sin(\angle FGH) &= \frac{\sin(26^\circ)}{47} \times 83 \\ \angle FGH &\approx 50.7^\circ\end{aligned}$$

But since  $\angle FGH$  is obtuse,

$$\angle FGH \approx 180^\circ - 50.7^\circ = 129.3^\circ$$

$$\angle GFH = 180^\circ - 129.3^\circ - 26^\circ = 24.7^\circ$$

$$\begin{aligned}\frac{GH}{\sin(24.7^\circ)} &= \frac{47}{\sin(26^\circ)} \\ GH &= \frac{47}{\sin(26^\circ)} \times \sin(24.7^\circ) \\ &\approx 44.80\end{aligned}$$

### Question 2

The area of  $\triangle PMN$  is

$$A_{\triangle PMN} = \frac{1}{2} \times 4 \times 8 \times \sin(110^\circ) \approx 15.04\text{m}^2$$

The length of  $PN$

$$\begin{aligned}PN^2 &= 4^2 + 8^2 - 2 \times 4 \times 8 \times \cos(110^\circ) \\ PN &= \sqrt{101.889...} \\ &\approx 10.1\text{m}\end{aligned}$$

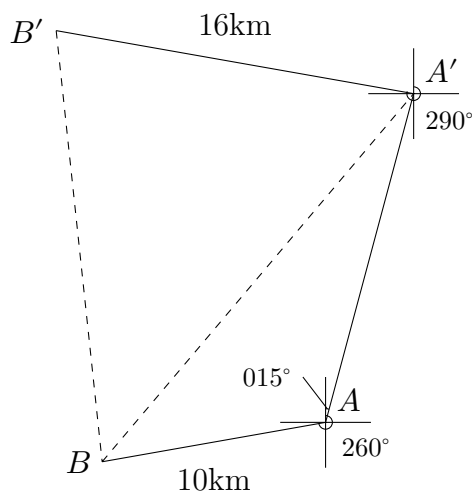
The area of equilateral  $\triangle PNQ$

$$A_{\triangle PNQ} = \frac{1}{2} \times 10.1^2 \times \sin(60^\circ) \approx 44.17\text{m}^2$$

The cost is

$$(15.04 + 44.17) \times 45 = \$2664.45$$

### Question 3



The length of  $AA'$  is  $5 \times 3 = 15$ .

Find the angle  $\angle BAA'$ ,  $360^\circ - 260^\circ + 15^\circ = 115^\circ$ .

Use the cosine rule to find the length of  $A'B$ ,

$$\begin{aligned} A'B &= \sqrt{10^2 + 15^2 - 2 \times 10 \times 15 \times \cos(115^\circ)} \\ &\approx 21.26 \end{aligned}$$

Find  $\angle BA'A$  using cosine rule

$$\begin{aligned} \angle BA'A &= \cos^{-1} \left( \frac{15^2 + 21.26^2 - 10^2}{2 \times 15 \times 21.26} \right) \\ &\approx 25.22^\circ \end{aligned}$$

Find the angle of  $B'A'B$  is  $290^\circ - 180^\circ - 15^\circ - 25.22^\circ = 69.78^\circ$

Use the cosine rule to find  $BB'$ ,

$$\begin{aligned} B'B &= \sqrt{16^2 + 21.26^2 - 2 \times 16 \times 21.26 \times \cos(69.78^\circ)} \\ &\approx 21.75 \end{aligned}$$

The speed of Ship 2 is  $21.75 \div 3 = 7.25\text{km/hr}$ .